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ORBITAL SENTINEL

Onboard AI Activity & Experiment-Protocol Validator

For: AI Human Activity Recognition for On-board BAS Experiments



- **Problem Statement ID –** 26174
- **Problem Statement Title-** AI Human Activity Recognition for On-board BAS Experiments
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-** 168345
- **Team Name (Registered on portal)-** NexGen Creators

Future astronauts cannot rely on real-time ground support to verify every experiment step. Orbital Sentinel validates the procedure onboard, detects missed or out-of-sequence actions, and provides immediate voice guidance - fully offline.

Live Activity Tracking	Next-Step Voice Prompt	Out-of-Sequence Alert	JSONL Log + recorded video	Local Video + IP Stream	Fully Offline

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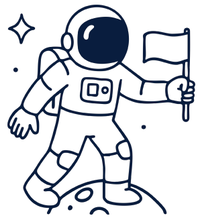
IDEA TITLE

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THE PROBLEM: Space stations run on restricted bandwidth — real-time ground support for experiments is physically impossible, and a missed step can ruin a non-repeatable science experiment.

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- Onboard AI Activity & Experiment-Protocol Validator
- Watches the astronaut via a fixed payload camera and validates activity against the known steps of a science experiment—entirely offline, at the edge.



TARGET EXPERIMENT: BAS-EXP-07 — Microbial Growth Assay

1. Retrieve sample vial
2. Open sample container
3. Insert pipette & draw sample
4. Transfer sample to growth medium
5. Seal container & attach sensor probe



3D — **Rack-relative tracking (V1, built)**
2D tracking vs. the payload rack, not gravity — see Slide 3

VOX — **Voice-first next-step + alerts**
Suggests next step; speaks up on any skip/out-of-order action

OFF — **Offline + local video + IP stream**
Zero ground link needed; records locally and streams to any set IP

System continuously suggests the next step and alerts (voice) on any skip or wrong order.

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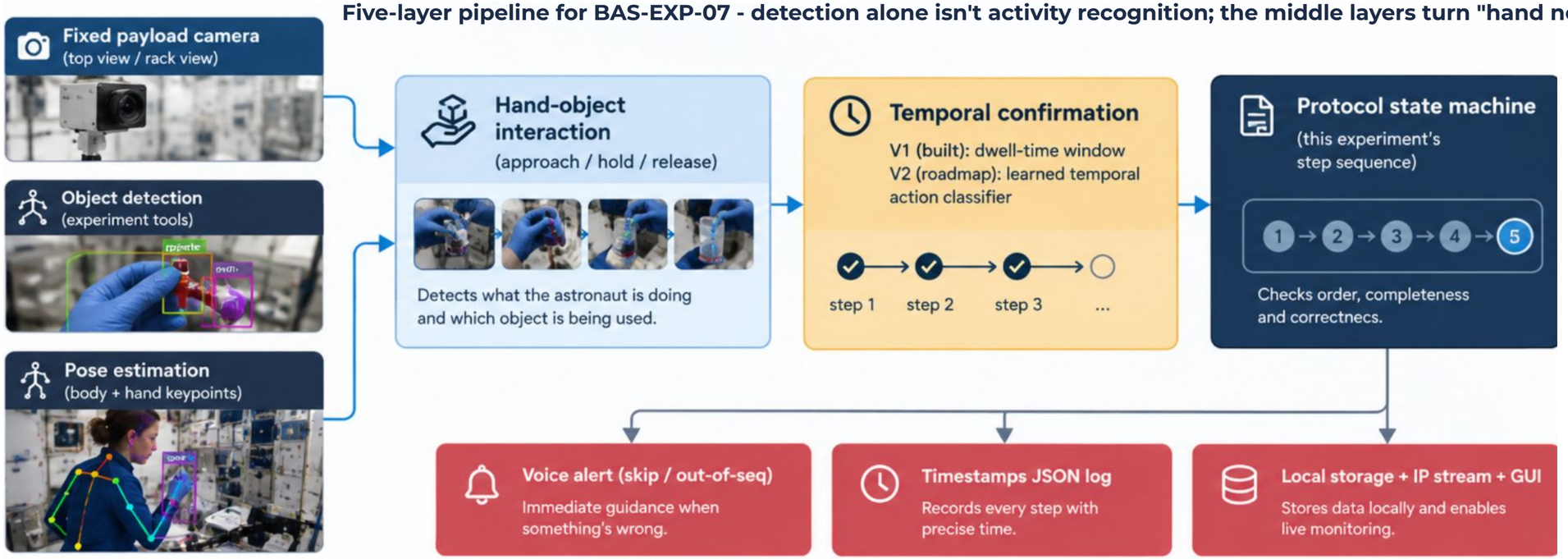
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TECHNICAL APPROACH

- Tech stack: Python . MediaPipe / YOLOv8-Pose . YOLOv8 . ArUco (OpenCV) . pyttsx3 (offline TTS) . JSON Lines . PyQt5 . OpenCV VideoWriter + lightweight IP streamer . Jetson-class target
- ACTIVITY = Pose + Object + Hand Object Interaction + Rack Zone + Temporal Context

Five-layer pipeline for BAS-EXP-07 - detection alone isn't activity recognition; the middle layers turn "hand near object" into "step 3 completed":



Measured on dev hardware:
~49.5 FPS pose-only; ~8.5 FPS combined with object detection (dev CPU, no GPU)
A step also requires the correct tool detected in-hand (containment), not merely near the zone.

A step is marked complete only after dwell-time confirmation — a single noisy frame cannot trigger a false advance.

Built by team: rack-relative homography, dwell-time confirmation, a real YOLOv8 object detector (demo mode placeholder classes), hand-object containment wired into the protocol matcher (24/24 tests pass), voice alerts, JSONL logging, local MP4 + IP stream, GUI. Open-source: MediaPipe/YOLOv8 weights (fine-tuning on real tools is next).

Low-bandwidth mode: video can be compressed (reduced resolution/frame rate, efficient encoding) for downlink to mission control; full-quality video always stays stored locally on-device.

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FEASIBILITY AND VIABILITY

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- Feasible today: pose/object detection and offline TTS are mature open-source components. The real engineering effort is the rack-relative tracking and the training dataset.

Future extension (not built in this version): the same camera + zone-checking + voice-alert pipeline could later extend to selected astronaut daily-life safety activities.

Actually validated so far:

24/24 automated tests pass (incl. object-aware containment logic)
Real YOLOv8 object detection + hand-object containment wired into the live matcher

~49.5 FPS pose-only / ~8.5 FPS combined with object detection (dev CPU, no GPU)
Local video recording + IP stream pipeline working
Working desktop GUI + JSON logs - real screenshots (Slide 6)
Out-of-sequence detection triggers correctly (Slide 6)

Real-tool detector - runs today on placeholder classes; fine-tuning swap is a one-flag change

Measured, not guessed - not yet claimed:

Activity accuracy / precision / recall / F1 - withheld until measured on real data.will be reported after BAS-specific validation.

Real BAS-EXP-07 dataset - collection protocol defined;
recording is the immediate next step

Jetson/edge benchmark - dev-CPU numbers shown now;
hardware test is scheduled next

Full 3D orientation-agnostic HMR - scoped as V2 (V1 = full protocol pipeline)

Risk	Mitigation strategy
No real dataset exists yet for training	Self-recorded analog dataset planned from a mock BAS-EXP-07 rig, annotated with a hierarchical protocol->step->action schema
In Gravity Object detector not yet fine-tuned on real tools	Ship with dwell-time zone tracking now (v1); fine-tune YOLOv8 once experiment hardware/tools are photographed
False alerts causing alert fatigue	Dwell-time confirmation (built + tested) - a single noisy frame cannot trigger a false alert

Scalability: the same offline procedural-monitoring pipeline applies to pharmaceutical cleanroom protocols, surgical checklist verification, and hazardous industrial maintenance on Earth.

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IMPACT AND BENEFITS

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Impact on target users (astronauts & mission science teams) - before vs. after:



~49.5 FPS

pose-only loop, dev CPU (no GPU);
with object detection added: ~8.5 FPS

Withheld until measured

voice-alert latency - measurement scheduled before finals, not guessed

24 / 24

automated logic tests passing
(pytest - incl. object-aware containment)

How this compares - V1 scope stated honestly:

Approach	Offline	Rack-relative tracking (2D)	Voice alerts + next-step	Video store + IP stream
Manual paper/checklist protocol	Yes	N/A	No	No
Ground-based HAR / pose models	No	No	No	Usually cloud
Orbital Sentinel - V1 (built)	**Yes**	**Yes**	**Yes**	**Yes**

Full 3D orientation-agnostic tracking (true microgravity) is V2 roadmap - not yet built. Stated once, honestly, here.

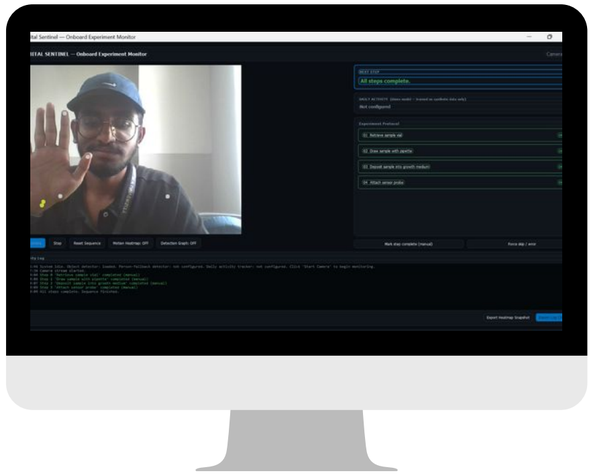
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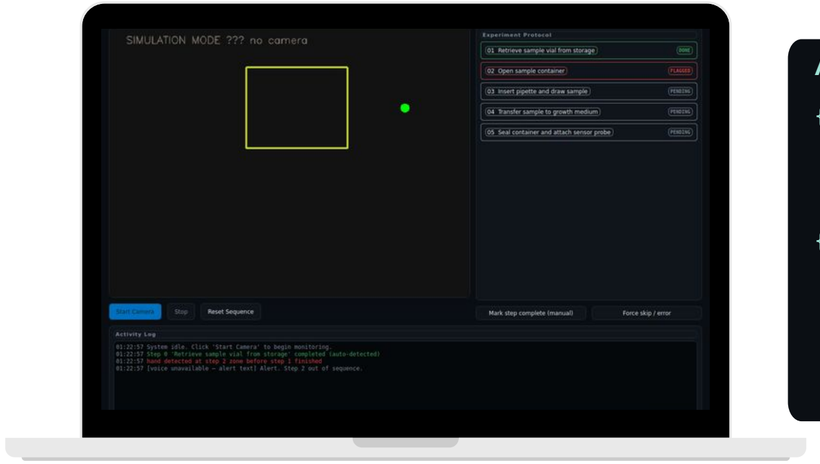
PROTOTYPE EVIDENCE

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- **Real prototype evidence - live camera capture + logic validation (see captions):**



"Real webcam + live AI perception with automatic activity-to-protocol validation."



Scripted run isolating the out-of-sequence decision logic

Actual log output:

```
{"step_id": 0, "expected_step":
  "Retrieve sample vial...", "status": "OK"}

{"step_id": 1, "expected_step":
  "Open sample container",
 "detected_action": "out-of-sequence action",
 "status": "ERROR"}
```

Next-step example: GUI: "Next: Insert pipette & draw sample" *Voice: "Step completed. Next: Insert pipette and draw sample."*

Every run produces: (1) timestamped JSONL log (2) local MP4 video file (3) live stream available at the configured IP (receiver script included).

Research references: *(Assembly101 — Sener et al., CVPR 2022 — procedural activity & action understanding,MediaPipe Hands — Zhang et al., 2020 — real-time hand landmark tracking,YOLOv8 — Ultralytics, 2023 — real-time object detection & pose estimation,OpenPose — Cao et al., 2018 — human pose estimation,ProMQA-Assembly — Hasegawa et al., 2025 — procedural task & action understanding,ArUco — Garrido-Jurado et al., Pattern Recognition, 2014 — fiducial marker-based spatial tracking,IndustReal — procedural assembly activity understanding and error detection.)*

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